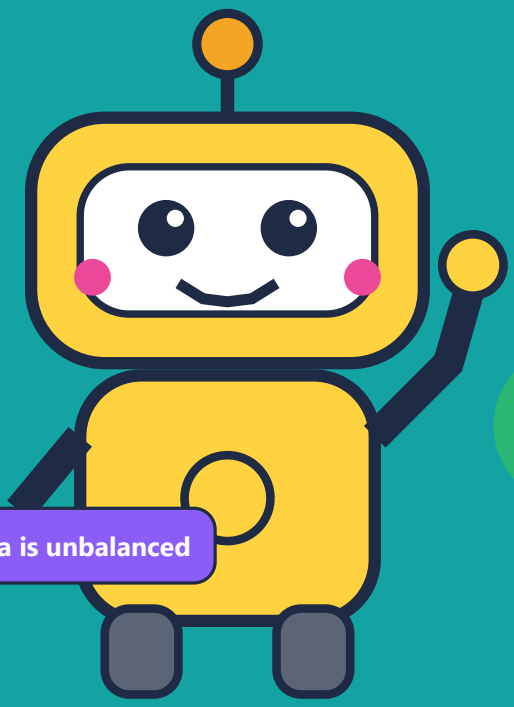
CodeSchool · Class 8

Think Like a Coder

Computational thinking and AI for Class 8

Learn it · See it · Try it · Build it



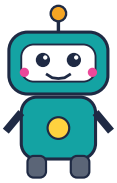
Follows NCERT Class 8 maths, CBSE's four AI chapters and two projects

Built for CBSE's Computational Thinking and AI curriculum, 2026-27

Theory, worked examples, practice, coding on paper, AI and projects

SAMPLE EDITION · DRAFT

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Welcome to Class 8! I am Robo. This year we write numbers in any base, sort shapes with if-else rules, calculate like a computer, run a full AI project from problem to model, make AI fairer, and lead two real projects.

How this book works

Learn	Robo explains one big idea in simple words, with the new coder words to remember.
Example	A worked example shows the idea step by step.
Try it!	Five practice questions. Circle the answer or write it in the box.
Now you build it	Coding on paper: put code cards in order, fix bugs and write your own programs.
Challenge	A harder puzzle for when you are ready for more.
How did I do?	Colour a face to show how sure you feel.
Projects	Two 20-hour team projects: plan, collect real data, build, test, present and score your work with a rubric.

My code cards

Coders use cards like these. Each colour means something different. You will find cut-out cards at the back of the book.

Step: do this **Repeat: a loop** **If: a decision** **Define: a function**

When: an event

For teachers and parents

Chapters 1-7 go with the same chapters of the NCERT Class 8 maths textbook, so they can be used in maths periods. AI chapters 1-4 follow the AI part of CBSE's Class 8 handbook, and the two projects run across the year. CBSE sets 100 hours a year in Class 8: about 40 for computational thinking, 20 for AI and 40 for two projects. The coding here needs no computer: students write, trace and debug programs on paper, which prepares them for block and text coding. Answers are at the back. Encourage children to explain their thinking, not just the answer.

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What's inside

1. A Square and a Cube 4
2. Power Play 7
3. A Story of Numbers 10
4. Quadrilaterals 13
5. Number Play 16
6. We Distribute Yet Things Multiply 19
7. Proportional Reasoning 22
AI 1. AI Project Lifecycle 25
AI 2. AI and Its Applications 28
AI 3. Data and Fairness in AI 31
AI 4. Ethics and Responsible AI 34
Project 1. Sports for Everyone 37
Project 2. Fair AI Audit 40
Answers 43
Coder words 45
Cut-out code cards 47

Coder skills in this book

Levels: C6 squares, powers, number bases and shape classifiers; C7 divisibility, identities and proportion; C8 AI projects, fairness and ethics.

Ch	Coder skill	Level	Ch	Coder skill	Level
1	Test squares and cubes with prime factors and loops	C6	AI1	Define, collect, train, test and deploy	C8
2	Exponents, doubling and counting combinations	C6	AI2	AI for people and planet; training no-code models	C8
3	Number bases: binary, ternary and changing base	C6	AI3	Spot unbalanced data and rebalance it	C8
4	Classify shapes with if-else rules	C6	AI4	Privacy, checking facts and human accountability	C8
5	Divisibility rules and remainders as quick tests	C7	P1	Survey, analyse with ratios and recommend changes	C8
6	The distributive property and identities for fast calculation	C7	P2	Test a classifier for bias, rebalance and report	C8
7	Ratios, the unitary method and scaling	C7			

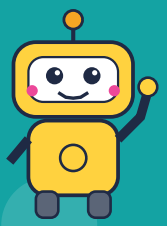
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CHAPTER 1 · LEVEL C6

1 A Square and a Cube

Coder skill: Test squares and cubes with prime factors and loops

Goes with NCERT maths, Class 8, chapter 1

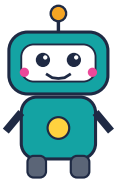


Learn

A **perfect square** is a number times itself (49 = 7 x 7); a **perfect cube** is a number used three times (64 = 4 x 4 x 4). Write a number as a product of primes: it is a square if the primes **pair up**, and a cube if they make **groups of three**.

A computer can also test 'is n a square?' with a loop: try k = 1, 2, 3, ... until k x k reaches n. Squares and cubes also hide patterns, like 1 + 3 + 5 + ... + (2n - 1) = n x n.

perfect square a number times itself	**perfect cube** a number used three times	**prime factorisation** a number as a product of primes	**square root** the number that was squared



Is 1728 a perfect cube? Break it into prime factors. If they make groups of three, it is!

Example

Is 1728 a perfect cube?

1. Prime factors: 1728 = 2 x 2 x 2 x 2 x 2 x 2 x 3 x 3 x 3.
2. Make groups of three: (2 x 2 x 2) (2 x 2 x 2) (3 x 3 x 3). Nothing is left over, so it is a cube.
3. Take one number from each group: 2 x 2 x 3 = **12**, so 1728 = 12 x 12 x 12.

★ *Coder fact!* Srinivasa Ramanujan noticed that 1729 is the smallest number that is a sum of two cubes in two ways: $1^3 + 12^3$ and $9^3 + 10^3$.

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Try it!

Circle the right answer, or write it in the box.

1 How many perfect squares are there between 50 and 150?

2 What is the smallest number you must multiply 72 by to get a perfect square?

a) 2 **b)** 3 **c)** 6 **d)** 8

3 56 x k is a perfect cube. What is the smallest whole number k?

a) 7 **b)** 14 **c)** 49 **d)** 343

4 Which 2-digit number is both a perfect square and a perfect cube?

a) 16 **b)** 27 **c)** 36 **d)** 64

5 A perfect square can never end in which digit?

a) 1 **b)** 4 **c)** 7 **d)** 9

Now you build it

An algorithm that tests whether n is a perfect square.

Input n | **Set k to 1** | **Repeat until k x k is at least n** | **change k by 1**

If k x k = n | **say 'square', else say 'not a square'**

Trace it for n = 49 and n = 50. Where does k stop, and how many times does the loop run?

Challenge

$2^3 - 1^3 = 7$, $3^3 - 2^3 = 19$, $4^3 - 3^3 = 37$. Find $13^3 - 12^3$ without working out the cubes. Explain your rule.

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✓ How did I do?

I can test squares and cubes with prime factors.	I can trace a checking algorithm.	I can find patterns in squares and cubes.
		

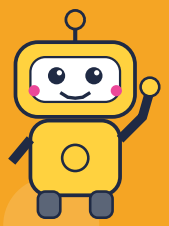
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CHAPTER 2 · LEVEL C6

2 Power Play

Coder skill: Exponents, doubling and counting combinations

Goes with NCERT maths, Class 8, chapter 2

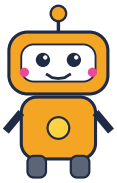


Learn

An **exponent** says how many times the **base** is multiplied: $2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$. When multiplying powers of the same base, add the exponents: $2^3 \times 2^4 = 2^7$.

Powers also count **combinations**. If you make several choices one after another, multiply the number of options: 3 shirts and 4 trousers give 12 outfits. With n yes/no choices there are 2^n combinations, which is why every extra character makes a password much harder to guess.

exponent	base	power	combination
how many times to multiply	the number being multiplied	a base with an exponent	one way of choosing



Fold a paper 10 times and it has 2 x 2 x … x 2 = 1,024 layers. Powers grow fast!

Example

Folding a sheet of paper

1. Each fold **doubles** the number of layers.
2. After n folds there are 2^n layers.
3. After 10 folds: 2^{10} = **1,024** layers, thicker than a 400-page book.

Folds	Layers	As a power
1	2	2^1
2	4	2^2
3	8	2^3
10	1,024	2^{10}

★ *Coder fact!* On the chessboard of the story, doubling grains on all 64 squares would need more than 18 quintillion grains in all.

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Try it!

Circle the right answer, or write it in the box.

1 $2^5 \times 2^3 = 2^{?}$

2 What comes next? 4, 10, 28, 82, ?

a) 244 **b)** 246 **c)** 162 **d)** 164

3 You have 4 shirts and 3 pairs of trousers. How many different outfits can you make?

4 How many different 4-digit PINs can be made with the digits 0 to 9?

a) 40 **b)** 1,000 **c)** 10,000 **d)** 9,999

5 Which is bigger: 2^{10} or 10^3?

a) 2^{10} **b)** 10^3

c) equal

Now you build it

The chessboard story: 1 grain on the first square, then double on each square.

Set grains to 1 | Set total to 0 | Repeat 10 times | change total by grains | set grains to grains x 2 | say total

Trace it for the first 10 squares. What does it say? Can you see a shortcut?

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★ Challenge

A team of 3 is chosen from 5 students A, B, C, D and E, but A and B cannot both be in it. How many teams are possible?

✓ How did I do?

I can use the laws of exponents.	I can count combinations by multiplying choices.	I can explain why doubling grows so fast.
		

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CHAPTER 3 · LEVEL C6

3 A Story of Numbers

Coder skill: Number bases: binary, ternary and changing base

Goes with NCERT maths, Class 8, chapter 3



Learn

In a **place-value** system each place is worth the **base** times the place to its right. Base 10 uses digits 0-9 and places 1, 10, 100, ... Base 2 (binary) uses 0 and 1 and places 1, 2, 4, 8, ... Base 3 (ternary) uses 0, 1 and 2 and places 1, 3, 9, 27, ...

Any number can be written in any base. To convert, divide by the base again and again and read the remainders backwards. Older systems like Roman numerals have no place value and no zero, which makes calculating much harder.

number base how many digits a system uses	**place value** what a position is worth	**ternary** base 3	**numeral** a symbol for a number

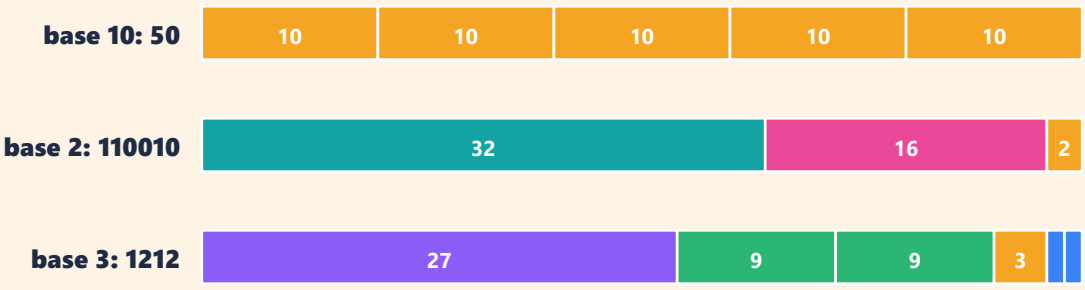


Measure 50 inches with pipes! Decimal pipes are 1, 10, 100 long. Binary pipes are 1, 2, 4, 8, 16, 32. Ternary pipes are 1, 3, 9, 27.

Example

50 inches with three kinds of pipe

1. Decimal: 50 = 5 tens, so **50**.
2. Binary: 50 = 32 + 16 + 2, so **110010**.
3. Ternary: 50 = 1 x 27 + 2 x 9 + 1 x 3 + 2 x 1, so **1212**. The same length, three different names.



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Try it!

Circle the right answer, or write it in the box.

1 What is 10 in binary?

a) 1010 **b)** 1111 **c)** 1000 **d)** 1100

2 What is the largest digit allowed in base 5?

a) 5 **b)** 4 **c)** 9 **d)** 1

3 (1021) in base 3 is which decimal number?

4 How is 1947 written in Roman numerals?

a) MCMXLVII **b)** MCMXXXXVII **c)** MDCCCCXLVII **d)** MCMLXVII

5 With binary pipes (one each of 1, 2, 4, 8, 16, 32), how many pipes measure 63 inches?

a) 2 **b)** 5 **c)** 6 **d)** 7

Now you build it

One algorithm that converts to any base.

Input n and base b **Repeat until n is 0** **write down the remainder of n ÷ b**

set n to n ÷ b, dropping the remainder **read the remainders from last to first**

Trace it for n = 100 and b = 3. Write each n and each remainder.

Challenge

Why do computers use binary rather than base 10? Then: what is the largest number you can write with 4 ternary digits?

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✓ How did I do?

I can write numbers in base 2, 3 and 10.	I can convert using repeated division.	I can compare place value with Roman numerals.
		

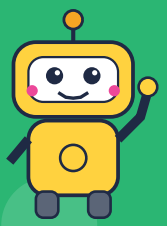
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CHAPTER 4 · LEVEL C6

4 Quadrilaterals

Coder skill: Classify shapes with if-else rules

Goes with NCERT maths, Class 8, chapter 4

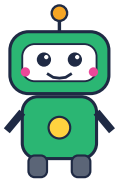


Learn

Quadrilaterals form a family. A **trapezium** has one pair of parallel sides; a **parallelogram** has two. A parallelogram with right angles is a **rectangle**; with all sides equal it is a **rhombus**; with both it is a **square**. A **kite** has two pairs of equal sides next to each other.

The angles of any quadrilateral add up to **360°**. Because shapes are defined by properties, we can sort them with a chain of if-else questions, the same way a decision tree classifies data.

parallelogram	rhombus	trapezium	diagonal
two pairs of parallel sides	four equal sides	one pair of parallel sides	joins opposite corners



I sort shapes like an AI classifier: ask about parallel sides, then right angles, then equal sides!

Example

Properties decide the name

1. Two pairs of parallel sides puts a shape in the parallelogram family.
2. Add right angles: rectangle. Add equal sides: rhombus. Add both: square.
3. So every square is also a rectangle and a rhombus, but not every rectangle is a square.

Shape	Parallel sides	Diagonals
Square	2 pairs	equal; bisect at 90°
Rectangle	2 pairs	equal; bisect each other
Rhombus	2 pairs	bisect at 90°
Parallelogram	2 pairs	bisect each other
Trapezium	1 pair	no special rule
Kite	none	one bisects the other at 90°

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Try it!

Circle the right answer, or write it in the box.

1 Which of these is always a parallelogram?

a) kite **b)** trapezium

c) rhombus **d)** any quadrilateral

2 Three angles of a quadrilateral are 80°, 95° and 110°. What is the fourth?

3 Which statement is true?

a) Every square is a rectangle **b)** Every rectangle is a square

c) Every rhombus is a square **d)** Every kite is a parallelogram

4 One angle of a parallelogram is 65°. What is the angle next to it?

5 How many rectangles (including squares) can you find in a 2 by 2 grid of squares?

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Now you build it

A quadrilateral classifier written with if-else.

Input a quadrilateral

If it has 2 pairs of parallel sides

If all angles are 90°

all sides equal? 'square', else 'rectangle'

else: all sides equal? 'rhombus', else 'parallelogram'

else: 1 pair parallel? 'trapezium', else 'other'

Run it for: (a) 4 equal sides with angles 60° and 120°; (b) only one pair of parallel sides; (c) opposite sides equal and all angles 90°.

Challenge

The diagonals of a quadrilateral are equal and cut each other in half at right angles. What shape must it be? Explain.

How did I do?

I can sort quadrilaterals by their properties.	I can use the angle sum of 360°.	I can write a classifier with if-else rules.
		

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CHAPTER 5 · LEVEL C7

5 Number Play

Coder skill: Divisibility rules and remainders as quick tests

Goes with NCERT maths, Class 8, chapter 5

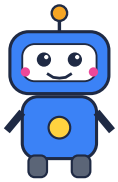


Learn

Divisibility rules test a number without long division: by 3 or 9, add the digits; by 4, look at the last two digits; by 8, the last three; by 11, find the alternating sum of the digits (+, -, +, ...).

The **remainder** is what is left after dividing. Remainders repeat in cycles, like the days of the week: 100 days after Monday is the same weekday as 2 days after Monday, because 100 leaves a remainder of 2 when divided by 7.

divisible	remainder	divisibility rule	multiple
divides with no remainder	what is left over	a shortcut test	in the times table



A divisibility rule is a shortcut algorithm: I can test 7,392 for 11 without dividing!

Example

Testing 7,392 without dividing

1. Digit sum 7 + 3 + 9 + 2 = 21: divisible by 3, but not by 9.
2. Last two digits 92 = 4 x 23: divisible by 4. Last three 392 = 8 x 49: divisible by 8.
3. Alternating sum 7 - 3 + 9 - 2 = 11: divisible by **11**. Check: 7,392 = 11 x 672.

★ *Coder fact!* Book ISBNs end in a check digit worked out with remainders, so a scanner can catch a mistyped number.

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Try it!

Circle the right answer, or write it in the box.

1 Which number is divisible by 9?

a) 4,527 **b)** 3,862 **c)** 2,915 **d)** 7,604

2 What is the remainder when $2^{10} = 1{,}024$ is divided by 7?

3 A lamp glows when the number entered is divisible by 6. Which number makes it glow?

a) 46 **b)** 54 **c)** 64 **d)** 74

4 A number divisible by both 4 and 6 must also be divisible by ...

a) 10 **b)** 12 **c)** 24 **d)** only 2

5 Today is Monday. What day will it be 100 days from today?

a) Tuesday **b)** Wednesday **c)** Thursday **d)** Monday

Now you build it

The divisibility test for 11 as an algorithm.

Input a number **Set total to 0 and sign to +** **Repeat for each digit from the left**

add sign x digit to total, then flip the sign **If total is divisible by 11**

say 'divisible by 11'

Trace it for 9,174 and for 5,213.

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★ Challenge

Find the smallest number that leaves a remainder of 1 when divided by 2, 3, 4, 5 or 6, and is divisible by 7.

✓ How did I do?

I can use divisibility rules.	I can work with remainders.	I can trace a divisibility algorithm.
		

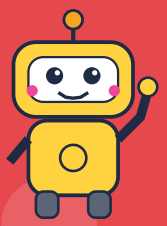
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CHAPTER 6 · LEVEL C7

6 We Distribute Yet Things Multiply

Coder skill: The distributive property and identities for fast calculation

Goes with NCERT maths, Class 8, chapter 6

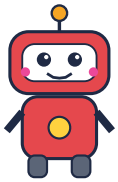


Learn

The **distributive property** says a(b + c) = ab + ac. With two brackets, every part multiplies every part: (a + b)(c + d) = ac + ad + bc + bd. An **area model** shows each part as a rectangle.

Some results are true for every number. These **identities** make calculation fast: $(a + b)^2 = a^2 + 2ab + b^2$, $(a - b)^2 = a^2 - 2ab + b^2$ and $(a + b)(a - b) = a^2 - b^2$. So $103 \times 97 = 100^2 - 3^2 = 9{,}991$. Computers use the same tricks to multiply huge numbers quickly.

distributive property	identity	expand	factorise
a(b + c) = ab + ac	true for every number	multiply out the brackets	write as a product



23 x 47 looks hard. Break it into (20 + 3) x (40 + 7) and I only need easy multiplications!

Example

23 x 47 with an area model

1. Split: 23 = 20 + 3 and 47 = 40 + 7.
2. Multiply each part: 800, 140, 120 and 21 (the model is not to scale).
3. Add: 800 + 140 + 120 + 21 = **1,081**.

	20	3
40	20 x 40 = 800	3 x 40 = 120
7	20 x 7 = 140	3 x 7 = 21

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Try it!

Circle the right answer, or write it in the box.

1 Use $(a + b)(a - b) = a^2 - b^2$ to work out 102 x 98.

2 Expand 3(2x + 5).

a) 6x + 15 **b)** 6x + 5 **c)** 5x + 8 **d)** 2x + 15

3 If a + b = 7 and ab = 10, what is $a^2 + b^2$?

4 Work out $47^2 - 43^2$ the quick way.

a) 360 **b)** 16 **c)** 400 **d)** 90

5 (x + 3)(x + 5) expands to ...

a) $x^2 + 8x + 15$ **b)** $x^2 + 15$ **c)** $x^2 + 8x + 8$ **d)** 2x + 8

Now you build it

A function that multiplies two numbers using squares.

```
Define times(a, b)
set m to (a + b) ÷ 2
set d to (a - b) ÷ 2
give back m x m - d x d
```

Use it for times(54, 46) and times(61, 59). Check one answer the long way.

Challenge

Why does times(a, b) work most easily when a and b are both even or both odd? Does it still work for 7 x 4?

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✓ How did I do?

I can use the distributive property.	I can use identities to calculate quickly.	I can expand two brackets with an area model.
		

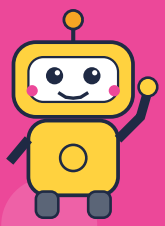
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CHAPTER 7 · LEVEL C7

7 Proportional Reasoning

Coder skill: Ratios, the unitary method and scaling

Goes with NCERT maths, Class 8, chapter 7

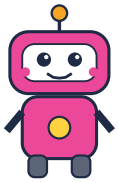


Learn

A **ratio** compares two quantities, like 3 : 5. Ratios are **in proportion** when they are equivalent: 3 : 5 = 12 : 20, because 3 x 20 = 5 x 12.

In **direct proportion** both grow together (more people, more rice). In **inverse proportion** one grows as the other shrinks (more painters, fewer days). The **unitary method** finds the value for one unit first, then scales up or down.

ratio	proportion	unitary method	inverse proportion
a comparison like 3 : 5	two equal ratios	find the value for one first	one up, the other down



To scale a recipe, I find the amount for one person, then multiply. It's the unitary method, and it's an algorithm!

Example

Direct and inverse proportion

1. Rice: 4 people need 300 g. One person needs 300 ÷ 4 = 75 g, so 10 people need 75 x 10 = **750 g** (direct).
2. Painting: 6 painters take 10 days, which is 60 painter-days of work.
3. 15 painters share the same 60 painter-days: 60 ÷ 15 = **4 days** (inverse).

Coder fact! Maps, scale models and recipes all rely on proportion: every part is scaled by the same ratio.

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Try it!

Circle the right answer, or write it in the box.

1 Simplify 36 : 48.

a) 3 : 4 **b)** 4 : 3 **c)** 6 : 8 **d)** 9 : 16

2 A map has a scale of 1 cm : 5 km. Two towns are 7.5 cm apart on the map. How far apart are they?

3 Three children's ages are in the ratio 2 : 3 : 4 and add up to 45. How old is the oldest?

4 8 workers finish a job in 12 days. How long would 6 workers take?

a) 9 days **b)** 16 days **c)** 10 days **d)** 18 days

5 Which ratio is in proportion with 3 : 5?

a) 12 : 20 **b)** 9 : 12 **c)** 5 : 3 **d)** 6 : 8

Now you build it

A function that scales a recipe.

```
Define scale(amount, from_people, to_people)
give back amount ÷ from_people x to_people
say scale(300, 4, 10) g of rice
say scale(2, 4, 10) cups of milk
```

What does the program say? Is a recipe an example of direct or inverse proportion?

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★ Challenge

A mixture has milk and water in the ratio 5 : 3 and holds 40 litres. How much water must be added to make the ratio 1 : 1?

✓ How did I do?

I can simplify and compare ratios.	I can use the unitary method.	I can tell direct from inverse proportion.
		

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AI1

AI CHAPTER 1 · LEVEL C8

AI Project Lifecycle

AI skill: Define, collect, train, test and deploy

Goes with the AI part of CBSE's Class 8 CT&AI handbook, chapter 1



Learn

An **AI project** builds a smart system that learns from data. Not every problem needs AI: a car wash that runs the same steps for every car is automation. The **AI project lifecycle** has five stages: (1) **define the problem**, (2) **collect and prepare data**, (3) **train a model**, (4) **evaluate and refine** it, and (5) **deploy** it for real use.

Raw data is messy. **Cleaning** removes missing values, duplicates, wrong values and irrelevant data. A model is tested on **test data** it has never seen, and its **accuracy** is the share of predictions it gets right.

problem statement what the AI must do	**data cleaning** fixing messy data	**model** what the AI learns from data	**deployment** putting it to real use



Building an AI is a project with stages. Get the problem and the data right first, or the model will be wrong!

Example

Cleaning a small dataset before training

1. The goal: predict whether a student will score more than 75%. Each row is one past student.
2. Rows 2 to 5 each have a problem. Fill or drop the missing value, delete the duplicate, check the impossible value and drop the irrelevant column.
3. Only clean, labelled data goes into training. Bad data in means bad predictions out.

Row	Data	Problem
1	Asha, 6 h study, 92%	OK
2	Ben, (blank), 81%	missing value
3	Asha, 6 h study, 92%	duplicate of row 1
4	Chetan, 25 h a day, 78%	impossible value
5	Dev, 4 h, 85%, likes blue	'likes blue' is irrelevant

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Try it!

Circle the right answer, or write it in the box.

1 What is the first stage of an AI project?

a) define the problem
b) deploy the model
c) train the model
d) collect prizes

2 A model gets 18 of 24 test cases right. What is its accuracy?

3 Which of these does NOT need AI?

a) a car wash that runs the same steps for every car
b) spotting spam emails
c) reading handwriting

4 Why is test data kept apart from training data?

a) to check the model on data it has not seen
b) to make training faster
c) it is not needed

5 A dataset has the same row twice. Which step fixes this?

a) remove duplicates
b) fill missing values
c) label the data

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Now you build it

An online shop wants an AI that predicts whether a delivery will be late. Plan the project.

- **Define: predict 'late' or 'on time' for each order**
- **Collect: distance, weather, traffic and result of past orders**
- **Prepare: clean the data and label each order**
- **Train on 80% of the orders**
- **Test on the other 20%; work out accuracy**
- **If accuracy is too low, improve the data and retrain**

The model gets 34 of 40 test orders right. What is its accuracy? The shop needs 90%. Should it deploy the model?

Challenge

Plan an AI that spots fake news: write one line for each stage of the lifecycle.

How did I do?

I can name the stages of an AI project.	I can clean a small dataset.	I can measure accuracy and decide whether to deploy.
		

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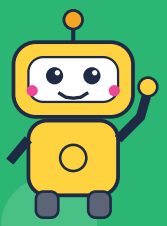
AI2

AI CHAPTER 2 · LEVEL C8

AI and Its Applications

AI skill: AI for people and planet; training no-code models

Goes with the AI part of CBSE's Class 8 CT&AI handbook, chapter 2

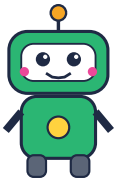


Learn

AI is helping across India: **satellite images** help find plastic dumping, **air-quality forecasts** warn people on bad days, **camera traps** count wildlife, **farm apps** diagnose crop problems from photos, and AI tools that read **chest X-rays** help screen for tuberculosis. Cities use AI for traffic, and schools for personalised learning.

No-code tools let anyone train a model: create **classes** (labels), add varied examples for each, train, then test with new examples. If the model fails, the fix is usually **better data**: more examples, more variety in background, light and voices.

no-code tool builds models without coding	**class** one label a model can choose	**sensor** a device that measures	**early warning** an alert before harm



With a no-code tool I can train my own model: give it labelled examples, train, test, and improve the data!

Example

Training a waste-sorting model with a no-code tool

1. Make three classes: plastic, paper and organic. Take 20-30 photos for each, with different backgrounds and light.
2. Upload and label the photos, then click train.
3. Test with items the model has never seen. Note every mistake, add better photos and train again.

water bottle	old newspaper	banana peel	chips packet
plastic	paper	organic	?

★ *Coder fact!* In India, AI tools that read chest X-rays are being used to help screen people for tuberculosis.

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Try it!

Circle the right answer, or write it in the box.

1 Finding plastic dumps in satellite images is AI helping ...

a) the environment **b)** banking **c)** sport

2 AI tools that read chest X-rays for signs of TB help ...

a) healthcare **b)** transport **c)** entertainment

3 A no-code AI tool lets you ...

a) train a model without writing code **b)** build a computer

c) skip collecting data

4 Your waste model works well in class but badly in the canteen. What is the most likely reason?

a) all its photos had the classroom background **b)** the canteen is too big

c) AI cannot see food

5 A voice-command model works for you but not for your grandmother. What should you do?

a) record samples from many different voices **b)** speak louder

c) delete the model

Now you build it

Plan a no-code voice-command model.

Classes: 'lights on', 'lights off', background noise

Record 15-20 samples of each: loud, soft, fast, slow **Train the model**

Test with new voices **If a class is often wrong, add more varied samples and retrain**

It was tested 30 times: 'lights on' 9 of 10 right, 'lights off' 6 of 10, noise 10 of 10. Which class needs more data? What is the overall accuracy?

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★ Challenge

Pick a problem in your school or town. Which kind of AI could help (vision, language or data), and what data would it need?

✓ How did I do?

I can give examples of AI helping society.	I can describe how a no-code model is trained.	I can improve a model by improving its data.
		

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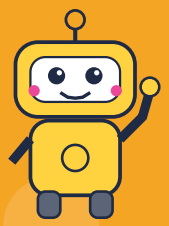
AI3

AI CHAPTER 3 · LEVEL C8

Data and Fairness in AI

AI skill: Spot unbalanced data and rebalance it

Goes with the AI part of CBSE's Class 8 CT&AI handbook, chapter 3

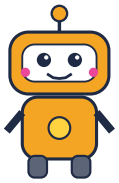


Learn

An AI looks for patterns in its **training data**. If a group is **over-represented** (too many examples) or **under-represented** (too few), predictions become unfair. A face detector trained mostly on one group may fail for others, and a translator trained on more 'male doctor' sentences may turn a gender-neutral Hindi sentence into 'He is a doctor'.

Fairness needs **representative** data: varied examples of every group and situation. Test each group separately, and **rebalance** the data by adding examples of the groups that are missing. Do not use features like gender to decide things they should not decide.

representative	over-represented	under-represented	rebalance
includes every group fairly	too many examples	too few examples	even out the data



I copy the patterns in my training data. If one group is missing or rare, I will be unfair to it, without even knowing!

Example

A sports-recommendation model (practice data)

1. A survey asked which sport students enjoyed. Most boys chose cricket and most girls chose badminton.
2. A model that learns from this would recommend badminton to every new girl, based on a pattern, not on her interest or ability.
3. Fairer: recommend from each student's own interests and skills, and do not use gender as a feature.

Sport	Boys	Girls
Cricket	48	14
Badminton	12	26

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Try it!

Circle the right answer, or write it in the box.

1 A model was trained on 90 photos of cricket bats and 10 of badminton rackets. Shown an unclear photo, what is it most likely to say?

a) cricket bat **b)** badminton racket

2 How should you fix the model in question 1?

a) add more varied racket photos **b)** add more bat photos

c) remove the racket photos

3 A translator turns a gender-neutral sentence into 'He is a doctor'. Why?

a) its training text had more examples of male doctors **b)** doctors are always men

c) the translator is broken

4 A face detector works badly for people with darker skin. Its training data was ...

a) not representative **b)** too large

c) too colourful

5 A dataset has 300 dog photos and 100 cat photos. What percentage are cats?

Now you build it

A rebalancing plan for an image dataset.

Count the examples in each class

Repeat until no class is less than half the biggest

collect more varied examples of the smallest class

Retrain and test each class separately

Classes: dogs 300, cats 100, rabbits 40. How many more cats and rabbits are needed so that no class is less than half the biggest?

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★ Challenge

Can a dataset be balanced in numbers but still unfair? Give an example.

✓ How did I do?

I can spot over- and under-represented data.	I can explain how unbalanced data leads to unfair results.	I can plan how to rebalance a dataset.
		

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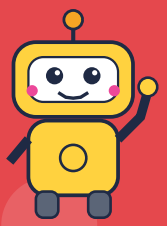
AI4

AI CHAPTER 4 · LEVEL C8

Ethics and Responsible AI

AI skill: Privacy, checking facts and human accountability

Goes with the AI part of CBSE's Class 8 CT&AI handbook, chapter 4

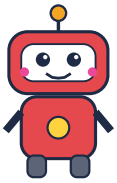


Learn

Responsible AI is fair, safe, transparent and good for society. **Privacy** means your personal data (name, location, photos, health) stays under your control: give **consent** only when an app really needs the data for its job.

AI can also spread **misinformation**, including fake images and videos, so check facts with trusted sources and read beyond headlines. **Accountability** means people, not machines, are responsible for decisions made with AI; in hospitals, banks and courts a human must review.

privacy control over your personal data	**consent** agreeing to share data	**misinformation** false information that spreads	**accountability** people answer for AI decisions



An AI can be fast and accurate and still be wrong to use. Responsible AI is fair, safe, open and respects your privacy!

Example

Which permissions should an app get?

1. Ask: does the app need this data to do its job?
2. If not, deny it. If yes, give the smallest permission that works, such as 'only while using the app'.
3. Check permissions again from time to time and remove ones you no longer need.

App asks for	Needed for its job?	Decision
Torch app: contacts	No	Deny
Maps app: location	Yes, while in use	Allow while using
Photo editor: photos	Only the chosen ones	Allow selected photos
Quiz game: microphone	No	Deny

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Try it!

Circle the right answer, or write it in the box.

1 A torch app asks to read your contacts. What should you do?

a) deny it **b)** allow it **c)** allow it once

2 A shocking video of a famous person may be fake. What should you do first?

a) check trusted news sources **b)** share it at once

c) comment on it

3 An AI in a hospital marks a patient as high risk. Who decides the treatment?

a) the doctor **b)** the AI

c) the patient's neighbour

4 Which of these is personal data?

a) your home address **b)** the capital of India

c) a maths formula

5 An AI system is fast and accurate for most people but often wrong for one group. Is it responsible to keep using it as it is?

a) no: it harms that group, so it must be fixed **b)** yes: most people are fine

Now you build it

A responsible-AI checklist written as rules.

Before using an AI tool | **If it asks for data it does not need** | **deny the permission**

If its answer affects a person | **a human checks before acting** | **If you share its output**

check the facts first

Use the checklist: an AI homework helper asks for your location and contacts, and gives you a 'fact' for your project.

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★ Challenge

If an AI system makes a harmful mistake, who should be responsible: the machine, its makers, or the people using it? Explain.

✓ How did I do?

I can protect my personal data.	I can check information before sharing.	I can explain why humans stay accountable for AI.
		

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P1

PROJECT 1 · 20 HOURS · LEVEL C8

Sports for Everyone

Project skill: Survey, analyse with ratios and recommend changes

CBSE: 40 project hours a year in Class 8, as two projects of 20 hours



The big question

Which students in our school play sport regularly, and what change could get more students active?

Your team runs an anonymous survey of at least 60 students: which class they are in, whether they play a sport at least three times a week, which sports, and, if not, why not. Ask permission first and collect no names.

Clean the data, work out percentages and ratios for each class, draw honest charts and look for patterns. Then recommend one or two changes, backed by your data, and plan how to test whether they work.

survey asking people the same questions	**anonymous** no names collected	**percentage** part ÷ whole x 100	**recommendation** advice backed by evidence

My project plan

Weeks	Stage	What I do	Done
1	**Define**	Agree the question and what 'plays regularly' means.	☐
2	**Design**	Write 5-6 clear survey questions. Plan how to ask permission and keep answers anonymous.	☐
3	**Collect**	Survey at least 60 students from different classes.	☐
4	**Clean**	Remove duplicates and blank forms. Note anything odd.	☐
5	**Analyse**	Work out percentages and ratios for each class, and the most common reasons.	☐
6	**Show**	Draw a bar chart for each class and a pie chart of reasons, with honest scales.	☐
7	**Find patterns**	Write three observations and one conclusion, with its limits.	☐
8	**Recommend**	Suggest one or two changes and how you would test them.	☐
9	**Present**	Present to your class or sports teacher in 3 minutes.	☐
10	**Reflect**	Score your work with the rubric and write what you would change.	☐

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How a coder does it

From survey to recommendation (practice data)

1. 80 students answered. Participation falls in the higher classes: 72% in Class 6 but only about 36% in Class 8.
2. The most common reason given in Class 8 was 'no time after school'.
3. Recommendation: a 20-minute lunchtime games slot for Class 8, then survey again after a month to see if participation rises.

Class	Play regularly	Percentage
6	18 of 25	72%
7	15 of 27	about 56%
8	10 of 28	about 36%

Survey tally sheet

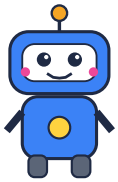
One row per class. Count answers with tallies first, then fill in the totals.

Class	Students asked	Play regularly	Percentage	Most common reason for not playing

My charts

Draw a bar chart of the percentage who play regularly in each class. Give it a title, labels and a scale that starts at 0.

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Data can change things! Collect it fairly, analyse it honestly, and let the evidence decide your recommendation.

★ How good is my project?

Score each row from 1 (just starting) to 4 (excellent). Then your teacher scores it too. Talk about any row where the scores differ.

What we check	What excellent (4) looks like	Me	Teacher
Question	A clear question about real participation.	1 2 3 4	1 2 3 4
Survey	Fair, anonymous questions; permission asked; 60 or more answers.	1 2 3 4	1 2 3 4
Analysis	Percentages and ratios correct; patterns found.	1 2 3 4	1 2 3 4
Charts	The right chart types, honest scales and clear labels.	1 2 3 4	1 2 3 4
Recommendation	Changes backed by the data, with a plan to test them.	1 2 3 4	1 2 3 4
Teamwork and talk	Work shared fairly; a clear presentation.	1 2 3 4	1 2 3 4

✓ Fair, safe and honest?

- [ ] We asked permission and explained how answers would be used.
- [ ] We collected no names, only class and answers.
- [ ] We did not recommend sports to anyone by gender.
- [ ] We reported results honestly, even surprising ones.

Looking back

What surprised you in your data?

How would you find out whether your recommendation worked?

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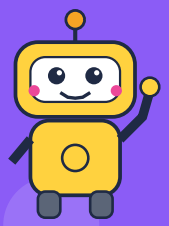
P2

PROJECT 2 · 20 HOURS · LEVEL C8

Fair AI Audit

Project skill: Test a classifier for bias, rebalance and report

CBSE: 40 project hours a year in Class 8, as two projects of 20 hours



The big question

Is a picture classifier equally good for everything it is used on, and can we make it fairer?

Your team audits a classifier: one you train with a no-code tool, like the cat-and-dog activity in CBSE's handbook, or the card version from your kit. Build a balanced test set it has never seen and measure the accuracy for each group.

Then study the training data: which groups or situations are missing or rare? Rebalance it, retrain, test again on the same test set, and write an audit report with your findings, your fix and the limits that remain.

audit a careful, independent check	**test set** examples kept back for checking	**group accuracy** accuracy for one group	**gap** difference between groups

My project plan

Weeks	Stage	What I do	Done
1	**Define**	Choose the classifier and the groups to compare (for example cats and dogs, indoor and outdoor).	☐
2	**Prepare**	Build a balanced test set: 20 unseen examples per group.	☐
3	**Test**	Run every test example and record the prediction and the true label.	☐
4	**Measure**	Work out the accuracy for each group and the gap between them.	☐
5	**Investigate**	Count the training examples per group. How are they alike, and how are they different?	☐
6	**Rebalance**	Add varied examples of the weak group: other backgrounds, lights and types.	☐
7	**Retest**	Retrain and test again on the same test set.	☐
8	**Compare**	Make a before-and-after table and a bar chart.	☐
9	**Report**	Write the audit report: findings, fix, remaining limits and advice.	☐
10	**Reflect**	Score your work with the rubric. Where could such gaps harm people?	☐

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How a coder does it

An audit in numbers (practice data)

1. Training data: 120 dog photos of many breeds, mostly outdoors; only 30 cat photos, mostly indoors.
2. Before: dogs 95% right but cats only 55%, a gap of 40 points, hidden inside 75% overall.
3. After adding 90 varied cat photos: dogs 90%, cats 85%. The gap shrank to 5 points and overall accuracy rose to 87.5%.

	Before	After
Dogs right	19 of 20 (95%)	18 of 20 (90%)
Cats right	11 of 20 (55%)	17 of 20 (85%)
Overall	30 of 40 (75%)	35 of 40 (87.5%)

Audit record

One row per test image. Write notes about anything that might matter, like background or light.

Image	Group	True label	Predicted	Right? (Y/N)	Notes

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An audit doesn't trust the overall score. Check every group, find the gap, fix the data, and test again!

★ How good is my project?

Score each row from 1 (just starting) to 4 (excellent). Then your teacher scores it too. Talk about any row where the scores differ.

What we check	What excellent (4) looks like	Me	Teacher
Audit question	Clear groups to compare and a fair, balanced test set.	1 2 3 4	1 2 3 4
Testing	Accuracy measured for each group on unseen data.	1 2 3 4	1 2 3 4
Investigation	Training data counted and its gaps explained.	1 2 3 4	1 2 3 4
Improvement	Data rebalanced and the model retested on the same test set.	1 2 3 4	1 2 3 4
Report	Findings, fix and remaining limits stated honestly.	1 2 3 4	1 2 3 4
Teamwork and talk	Work shared fairly; a clear presentation.	1 2 3 4	1 2 3 4

✓ Fair, safe and honest?

- [ ] We used only pictures we were allowed to use, with no photos of people.
- [ ] We tested every group with the same kind of test set.
- [ ] We reported the gap honestly, even after our fix.
- [ ] We said what our audit cannot show.

Looking back

What caused the biggest unfairness in your classifier?

Where in real life could a gap like this harm people?

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Answers

Many questions have more than one good answer. Accept any answer that follows the rule and can be explained.

1. A Square and a Cube

1. 5: 64, 81, 100, 121 and 144
2. a) 2 (72 = 2 x 2 x 2 x 3 x 3; 144 = 12 x 12)
3. c) 49 (56 = 2 x 2 x 2 x 7, so two more 7s are needed)
4. d) 64 = 8 x 8 = 4 x 4 x 4
5. c) 7 (squares end in 0, 1, 4, 5, 6 or 9)
Build it: 49: k stops at 7 after 6 loops and it says 'square'. 50: k stops at 8 after 7 loops; 64 is not 50, so 'not a square'
Challenge: 469: $n^3 - (n - 1)^3 = 3 \times n \times (n - 1) + 1$, and 3 x 13 x 12 + 1 = 469 (check: 2197 - 1728)

2. Power Play

1. 8
2. a) 244: each term is a power of 3 plus 1 ($3^5 + 1$)
3. 12
4. c) 10,000 (10^4)
5. a) 2^{10} = 1,024, against 1,000
Build it: 1 + 2 + 4 + ... + 512 = 1,023, which is $2^{10} - 1$
Challenge: 7: there are 10 teams of 3 in all, and 3 of them contain both A and B

3. A Story of Numbers

1. a) 1010
2. b) 4
3. 34 (27 + 0 + 6 + 1)
4. a) MCMXLVII
5. c) 6: 32 + 16 + 8 + 4 + 2 + 1
Build it: 100: rem 1, n = 33; rem 0, n = 11; rem 2, n = 3; rem 0, n = 1; rem 1, n = 0. Read backwards: 10201
Challenge: A switch has two states, on and off, so two digits are easy to store reliably. 2222 in base 3 = 54 + 18 + 6 + 2 = 80

4. Quadrilaterals

1. c) rhombus
2. 75°
3. a) every square is a rectangle
4. 115° (neighbouring angles add up to 180°)
5. 9: 4 small squares, 4 rectangles of 2 squares and 1 big square
Build it: (a) rhombus (b) trapezium (c) rectangle
Challenge: A square: diagonals that bisect each other make a parallelogram, equal diagonals make it a rectangle, and meeting at 90° makes it a rhombus

5. Number Play

1. a) 4,527 (digit sum 18)
2. 2 (7 x 146 = 1,022)
3. b) 54
4. b) 12, the lowest common multiple
5. b) Wednesday (100 = 14 x 7 + 2)
Build it: 9,174: 9 - 1 + 7 - 4 = 11, divisible (9,174 = 11 x 834). 5,213: 5 - 2 + 1 - 3 = 1, not divisible
Challenge: 301: numbers 1 more than a multiple of 60 are 61, 121, 181, 241, 301, ...; the first divisible by 7 is 301 = 7 x 43

6. We Distribute Yet Things Multiply

1. 9,996 (10,000 - 4)
2. a) 6x + 15
3. 29 (49 - 2 x 10)
4. a) 360 = (47 + 43)(47 - 43) = 90 x 4
5. a) $x^2 + 8x + 15$
Build it: 54 x 46: $50^2 - 4^2$ = 2,500 - 16 = 2,484. 61 x 59: $60^2 - 1^2$ = 3,599
Challenge: Otherwise (a + b) ÷ 2 is not a whole number. It still works with halves: $5.5^2 - 1.5^2$ = 30.25 - 2.25 = 28

7. Proportional Reasoning

1. a) 3 : 4
2. 37.5 km
3. 20 (10, 15 and 20)
4. b) 16 days (96 worker-days ÷ 6)
5. a) 12 : 20
Build it: 750 g of rice and 5 cups of milk; direct proportion
Challenge: 10 litres: there are 25 litres of milk and 15 of water, so 10 more litres of water are needed

AI 1. AI Project Lifecycle

1. a) define the problem
2. 75%
3. a) the car wash: it is automation
4. a) to check the model on data it has not seen
5. a) remove duplicates
Build it: 85%. Not yet: it should improve the data (for example more orders in bad weather) and retrain
Challenge: Open answer, for example: define 'real or fake headline'; collect checked real and fake headlines; clean and label them; train; test on new headlines; refine and only deploy with human checks

AI 2. AI and Its Applications

1. a) the environment
2. a) healthcare
3. a) train a model without writing code
4. a) its training photos all had the same background
5. a) train it with samples from many different voices
Build it: 'lights off' needs more varied samples; overall 25 of 30, about 83%
Challenge: Open answer, for example: a camera model that spots overflowing bins, trained on labelled photos of full and empty bins

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AI 3. Data and Fairness in AI

1. a) cricket bat, because that class dominates its training data
2. a) add more varied racket photos
3. a) its training data had more male doctors
4. a) not representative
5. 25%
Build it: Half of 300 is 150, so 50 more cats and 110 more rabbits
Challenge: Yes: 100 cat and 100 dog photos, but every cat is indoors and every dog outdoors, so the model learns the background; or all photos come from one region

AI 4. Ethics and Responsible AI

1. a) deny it: a torch does not need contacts
2. a) check trusted news sources
3. a) the doctor, after reviewing the AI's suggestion
4. a) your home address
5. a) no: it must be fixed for that group
Build it: Deny location and contacts; check the fact with a trusted source before using it, and say that you used AI
Challenge: People: the makers and the organisation using it are accountable, because a machine cannot be held responsible

Project 1. Sports for Everyone

Open project: assess with the rubric. Check percentages: part ÷ whole x 100, for example 18 ÷ 25 x 100 = 72%.

Project 2. Fair AI Audit

Open project: assess with the rubric. Group accuracy = right answers for that group ÷ test images in that group.

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Coder words

accountability	people answer for AI decisions	AI 4
anonymous	no names collected	Project 1
audit	a careful, independent check	Project 2
base	the number being multiplied	Ch 2
class	one label a model can choose	AI 2
combination	one way of choosing	Ch 2
consent	agreeing to share data	AI 4
data cleaning	fixing messy data	AI 1
deployment	putting it to real use	AI 1
diagonal	joins opposite corners	Ch 4
distributive property	a(b + c) = ab + ac	Ch 6
divisibility rule	a shortcut test	Ch 5
divisible	divides with no remainder	Ch 5
early warning	an alert before harm	AI 2
expand	multiply out the brackets	Ch 6
exponent	how many times to multiply	Ch 2
factorise	write as a product	Ch 6
gap	difference between groups	Project 2
group accuracy	accuracy for one group	Project 2
identity	true for every number	Ch 6
inverse proportion	one up, the other down	Ch 7
misinformation	false information that spreads	AI 4
model	what the AI learns from data	AI 1
multiple	in the times table	Ch 5
no-code tool	builds models without coding	AI 2
number base	how many digits a system uses	Ch 3
numeral	a symbol for a number	Ch 3
over-represented	too many examples	AI 3
parallelogram	two pairs of parallel sides	Ch 4
percentage	part ÷ whole x 100	Project 1
perfect cube	a number used three times	Ch 1
perfect square	a number times itself	Ch 1
place value	what a position is worth	Ch 3
power	a base with an exponent	Ch 2
prime factorisation	a number as a product of primes	Ch 1
privacy	control over your personal data	AI 4
problem statement	what the AI must do	AI 1

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proportion	two equal ratios	Ch 7
ratio	a comparison like 3 : 5	Ch 7
rebalance	even out the data	AI 3
recommendation	advice backed by evidence	Project 1
remainder	what is left over	Ch 5
representative	includes every group fairly	AI 3
rhombus	four equal sides	Ch 4
sensor	a device that measures	AI 2
square root	the number that was squared	Ch 1
survey	asking people the same questions	Project 1
ternary	base 3	Ch 3
test set	examples kept back for checking	Project 2
trapezium	one pair of parallel sides	Ch 4
under-represented	too few examples	AI 3
unitary method	find the value for one first	Ch 7

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Cut-out code cards

Ask an adult to help you cut along the dotted lines. Use the cards to build programs on your desk, then act them out with a friend.

Define ___	give back ___	Repeat until ___
Define ___	give back ___	Repeat until ___
Repeat for each ___	If ___	else
Repeat for each ___	If ___	else
AND	OR	Input ___
AND	OR	Input ___
Set ___ to ___	change ___ by ___	say ___
Set ___ to ___	change ___ by ___	say ___

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